MIL-C-47244(MI)
26 July 1974
SUPERSEDING
MIS 16062
6 December 1966

MILITARY SPECIFICATION

COATING, ABLATIVE RESISTANT

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two-component, room-temperature curing, filled, thermosetting, modified epoxy resin system that resists abrasion and erosion at extreme temperatures.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
TT-M-268	Methyl-Ethyl-Ketone, Technical
TT-S-735	Standard Test Fluids, Hydrocarbon

FSC 8030

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Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes
MIL-P-14232	Parts, Tools and Equipment for Army Material, Packaging and Packing of
MIL-P- 47215	Primer, Ablative Resistant Coating

STANDARDS

Federal

FED-STD-406	Plastics; Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Uniform Classification Committee

Uniform Freight Classification

Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60605.)

National Motor Freight Traffic Associations, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street N.W., Washington, D.C. 20036.)

Official Air Transport Restricted Articles Tariff No. 6-D

Title 49 of the Code of Federal Regulations (parts 170-189)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When a preproduction sample is required (see 6.2), it shall meet the requirements of this specification.

3.2 Material. The material shall be a two-component, room-temperature curing, filled, thermosetting, modified epoxy resin system. The resin system is formulated by mixing a base resin and a curing agent as instructed by the manufacturer.

3.3 Physical properties.

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3.3.1 Appearance. The cured resin system shall be continuous and uniform in texture. The color shall not vary from lot to lot.

3.3.2 Specific gravity. The specific gravity of the cured resin system shall be not less than 1.2 nor more than 1.6.

3.3.3 Application time. The application time of the uncured resin system shall be not less than 60 minutes at room temperature, defined as plus 25 plus or minus 3 degrees Celsius (C) (plus 77 plus or minus 5 degrees Fahrenheit (F)).

3.3.4 Flow. Flow of the freshly mixed resin system shall be not more than 0.3 inch when a cylindrical section approximately 3/8 inch high and 1 1/2 inch in diameter is allowed to flow under its own weight on a vertical surface. In addition, the resin system shall retain a flow not more than 0.3 inch throughout the entire application time.

3.3.5 Hardness. Hardness of the cured resin system shall be as specified in Table I.

Table I.
Hardness requirements

Curing conditions		Hardness value
Cond 1	Room temperature for 24 ± 2 hours	Not less than 40 Shore D
Cond 2	Room temperature for 7 days	Not less than 55 Shore D
Cond 3	Room temperature for 3 hours, followed by +66 ± 1 degrees C (+150 ± 2 degrees F) for 3 hours, followed by room temperature for not less than 3 hours.	Not less than 55 Shore D

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3.3.6 Low temperature stability. The cured resin system shall withstand exposure to a temperature of minus 54 plus or minus 3 degrees C (minus 65 plus or minus 5 degrees F) without cracking, checking, or loss of adhesion.

3.3.7 Humidity and heat resistance. The cured resin system shall withstand simultaneous exposure to a relative humidity of not less than 90 percent and temperature cycling as follows:

a. A temperature of plus 49 plus or minus 3 degrees C (plus 120 plus or minus 5 degrees F) for 3 days.

b. Return in not less than 3 hours to minus 12 plus or minus 3 degrees C (plus 10 plus or minus 5 degrees F) and maintained at this temperature for not less than 30 minutes.

c. Uncontrolled stabilization at room temperature.

d. Increase to plus 88 plus or minus 3 degrees C (plus 190 plus or minus 5 degrees F) in not less than 50 minutes, and maintained at this temperature for 4 hours.

No softening, sponging, blistering, checking, shrinking, or loss of adhesion shall occur. Weight of the resin system shall not increase more than 5 percent. Thickness of the resin system shall not increase or decrease more than 5 percent.

3.3.8 Tread resistance. The cured resin system, after continuous exposure for not less than 72 hours to a temperature of plus 88 plus or minus 3 degrees C (plus 190 plus or minus 5 degrees F), shall show no evidence of surface degradation or loss of adhesion as a result of foot traffic.

3.3.9 Low temperature impact resistance. The cured resin system, when stabilized at the temperature specified in 3.3.6, shall not shatter when impacted by a 1/2-pound steel ball dropped from heights up to 20 feet.

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3.3.10 Flammability. The cured resin system shall be self extinguishing.

3.3.11 Hydrocarbon resistance. The cured resin system shall not have any loss of adhesion or be adversely affected after exposure to hydrocarbons. The hydrocarbon-exposed resin system shall be self-extinguishing in not more than 2 seconds after ignition has been effected.

3.3.12 Blast erosion. The cured resin system shall withstand the erosive effects of a supersonic blast, with erosive loss limited to a thickness of 0.080 inch. Blast conditions shall be as follows:

a. Temperature and pressure above the surface of the resin system shall be not less than 2830 degrees C (5100 degrees F) and not less than 330 pounds per square inch gage (psig), respectively.

b. Direction of the blast shall be 90 plus or minus 5 degrees with respect to the surface of the resin system.

c. The chemical composition (by percentage weight plus or minus 5 percent) of the gas impinging on the resin system shall be:

(1) Steam - 1.01.

(2) Hydrogen - 3.47.

(3) Carbon monoxide - 36.88.

(4) Nitrogen - 7.94.

(5) Hydrogen chloride - 20.47.

(6) Aluminum oxide - 30.23.

d. Duration of blast shall be not less than 90 milliseconds.

3.3.13 Shelf life. The shelf life of the base resin and curing agent shall be not less than 6 months from the date

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of manufacture when stored in original unopened containers at a temperature less than plus 27 degrees C (plus 80 degrees F).

3.3.14 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform and in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of the resin system. Preproduction samples which do not meet the requirements of this specification shall be subject to rejection. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Classification of examinations and tests. The examination and testing of the resin system shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the resin system shall consist of the

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following examinations and tests:

a. Appearance.

b. Density.

c. Application time.

d. Flow.

e. Hardness.

4.3.2.1 Lot size. Lot size shall consist of all the resin system submitted for acceptance at the same time which has been prepared by the same company without a change in materials or processes in one continuous period of operation.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), 1 pint of resin and 1 pint of curing agent from one unit representative of each lot shall be selected at random for quality conformance testing. Each container of resin and curing agent shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Test conditions. Unless otherwise specified in the applicable test, all tests shall be performed at room temperature and at relative humidity between 40 and 60.

4.4.2 Test panels. Unless otherwise specified in the applicable test, test panels shall consist of 9 1/2-inch squares of 1/2-inch thick aluminum conforming to QQ-A-250/12 with the resin system applied to centered 5-inch squares on each aluminum square.

4.4.2.1 Precleaning. Prior to application of the resin system, the aluminum squares shall be cleaned by abrading the surface with 180 to 200 grit sandpaper, washed with methyl-ethyl-ketone solvent conforming to TT-M-261, and allowed to dry for not less than 20 minutes.

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4.4.2.2 Primer. A primer conforming to MIL-P- shall be brush-coated on the cleaned aluminum squares and allowed to air-dry for not less than 30 minutes.

4.4.2.3 Application of resin system. The quantity of resin system required for testing, not less than 90 grams of base resin, shall be mixed thoroughly in the proportion specified in 3.2 in not more than 5 minutes, preferably by a standardized mechanical mixing method. Unless otherwise specified, the test panels shall have the resin system troweled on the aluminum squares to a thickness of not less than 0.165 inch nor more than 0.205 inch and cured at room temperature for 7 days. Except for the hardness tests (see 4.4.3.5), the resin system shall be tested within 3 days after cure.

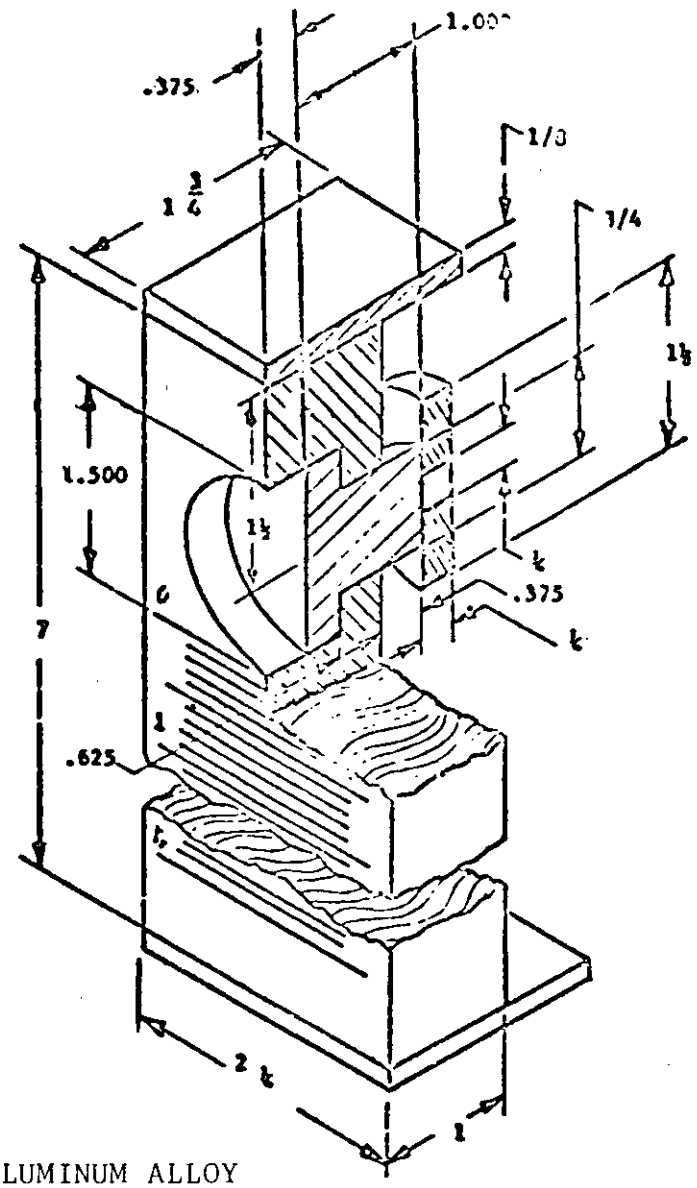
4.4.3 Physical property tests.

4.4.3.1 Appearance. The appearance of the resin system shall be checked visually and shall be as specified in 3.3.1.

4.4.3.2 Specific gravity. The specific gravity of the resin system shall be determined in accordance with FED-STD-406, Method 5011. The specific gravity shall be as specified in 3.3.2.

4.4.3.3 Application time. With the base resin, curing agent, and compound application gun stabilized at the conditions specified in 4.4.1 for not less than 8 hours, 90 grams or more of base resin shall be mixed thoroughly with the curing agent in the proportion specified in 3.2. The mixed resin system shall be promptly used to fill a standard 603 Semco compound gun cartridge having a Semco 254 nozzle with an orifice diameter of not less than 0.120 inch nor more than 0.130 inch. The gun and resin system shall be maintained at standard conditions throughout the test. The gun shall be attached to a constant supply of air or inert gas at not less than 85 pounds per square inch guage (psig) (for line variation or gage error). From 2 to 3 inches of the resin system shall be extruded initially to clear trapped air. At the end of 60 minutes, measured from the time mixing was begun, the resin system shall be extruded onto a suitable receptacle for 1 minute. The amount of resin system extruded during the 1-minute period shall be weighed and shall be not less than 20 grams (g).

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MATERIAL: ALUMINUM ALLOY

DIMENSIONS ARE IN INCHES. TOLERANCES:

THREE PLACE DECIMALS: ± 0.003
FRACTIONS: ± 1/32

Figure 1. Flow test jig

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4.4.3.4 Flow. The flow test shall include initial flow and flow at the end of the application time and shall be conducted with a test jig as shown in Figure 1. The flow test jig shall be placed on a table with the front face upward and with the plunger depressed to the limit of its travel. The application gun shall be filled and attached to the pressure sources as specified in 4.4.3.3. Within 15 minutes after the beginning of the mixing, enough of the mixed resin system shall be extruded from the application gun to fill the recessed cavity of the jig. The resin system shall not be worked with the gun tip or a spatula but shall be leveled off even with the block by scraping with a spatula in two passes, each starting in the center and moving toward the sides of the jig. Within 10 seconds after the leveling operation, the jig shall be placed on its base and the plunger immediately advanced to the limit of its forward travel. The initial flow measurement shall be taken not less than 25 nor more than 35 minutes after the resin system has been applied to the test jig. The flow shall be measured from the point tangent to the lower edge of the plunger to the farthest point to which flow has occurred. The measured flow shall be not more than 0.3 inch. The flow test shall be repeated not less than 55 nor more than 65 minutes after the beginning of the mixing of the resin system. The final flow measurement shall be taken not less than 25 nor more than 35 minutes after the resin system has been applied to the test jig.

4.4.3.5 Hardness. Hardness tests for the resin system shall be in accordance with FED-STD-406, Method 1083, except as follows.

4.4.3.5.1 Test panels. Test panels for the hardness tests shall be as specified in 4.4.2, except as follows:

a. Thickness of the cured resin system shall be not less than 0.150 inch nor more than 0.1875 inch.

b. The resin system shall be cured in accordance with conditions 1 and 2 of Table 1 for preproduction testing, and in accordance with condition 3 of Table 1 for quality conformance testing.

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4.4.3.5.2 Preproduction hardness test. Curing conditions and hardness values for preproduction testing shall be as specified in Table I for conditions 1 and 2.

4.4.3.5.3 Quality conformance hardness test. Curing conditions and hardness value for quality conformance testing shall be as specified in Table I for condition 3.

4.4.3.6 Low temperature stability. Two room-temperature test panels shall be placed in a temperature chamber for low temperature stability testing. The temperature shall be lowered to minus 54 plus or minus 3 degrees C (minus 65 plus or minus 5 degrees F) at a rate of 1 degree C (2 degrees F) per minute. The test panels shall remain at this temperature for not less than 1 hour, after which they shall be examined. There shall be no evidence of cracking, checking, or loss of adhesion.

4.4.3.7 Humidity and heat resistance. Two test panels shall be placed in a temperature-humidity chamber and subjected to the relative humidity level and temperature cycling as specified in 3.3.7. At the conclusion of the test period, the test panels shall be visually examined, weighed and measured to ensure compliance with the requirements of 3.3.7.

4.4.3.8 Tread resistance. One test panel shall be placed in a temperature chamber and shall be subjected to a temperature of plus 88 plus or minus 3 degrees C (plus 190 plus or minus 5 degrees F) for not less than 72 hours. The test panel shall be transferred directly from the temperature chamber to the platen of a Link spring checker, and a 3/4-inch thick by 3-inch diameter elastomer disk conforming to MIL-S-6855, class 2, grade 70 shall be positioned on the test panel as shown in Figure 2. A force of 200 plus or minus 10 pounds shall be applied to the disk. With the disk held rigid, the test panel shall be twisted approximately 45 degrees clockwise, then approximately 90 degrees counter-clockwise, followed by approximately 45 degrees clockwise. At the conclusion of the test, the resin system shall be examined and shall show no evidence of surface degradation or loss of adhesion.

4.4.3.9 Low temperature impact resistance. Two test panels shall be placed in a temperature chamber and stabilized at minus 54 plus or minus 3 degrees C (minus 65 plus or minus 5 degrees F). The test panels, while at this temperature,

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shall be tested for impact resistance in accordance with FED-STD-406, Method 1074. At the conclusion of the test, the resin system shall be examined and shall show no evidence of shattering or fracture.

4.4.3.10 Flammability. The cured resin system shall be tested for flammability in accordance with FED-STD-406, Method 2021. The cured resin system shall be nonburning by this test.

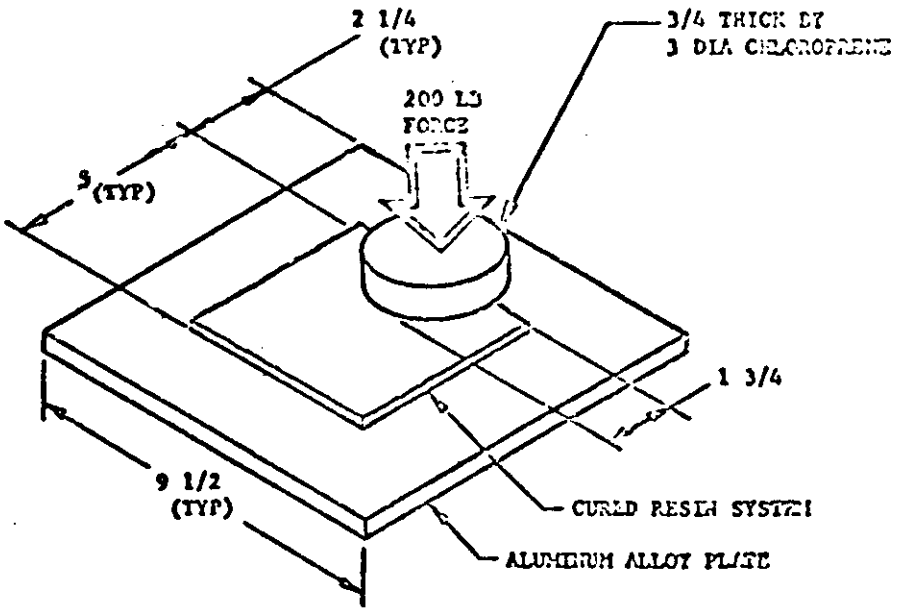
4.4.3.11 Hydrocarbon resistance.

4.4.3.11.1 Resistance to hydraulic oil. Two test panels shall be half immersed in hydraulic oil conforming to MIL-H-5606 for not less than 55 minutes nor more than 65 minutes. Upon removal, excess oil shall be wiped from the test panels. A standard 3/8-inch bunsen burner with air ports open to produce a blue flame approximately 1 inch high shall be placed under the test panel and adjusted so that the flame tip is just in contact with the portion of the resin system that had been immersed. At the end of 30 seconds the flame shall be removed and the test panel allowed to burn. In case the test panel does not continue to burn after the first ignition, the burner shall be placed under the test panel for a second period of 30 seconds immediately after the resin system ceases to burn. A stop watch shall be started when the flame is removed and the time elapsing before the resin system extinguishes itself shall be measured. Burning time shall be not more than 2 seconds.

4.4.3.11.2 Resistance to test fluid. The remaining half of each of the test panels shall be immersed in hydrocarbon test fluid conforming to TT-S-735, Type I, for not less than 25 minutes. Upon removal, the test panel shall be allowed to air-dry for not less than 30 minutes. The test specified in 4.4.3.11.1 shall be repeated except that the flame shall be applied to the portion of the resin system that had been immersed in the test fluid.

4.4.3.12 Blast erosion. The blast erosion test shall be performed with the use of a combustion chamber capable of generating exhaust products having the composition and characteristics specified in 3.3.12. The nozzle of the combustion chamber shall have an exit to throat area ratio

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DIMENSIONS ARE IN INCHES

TOLERANCES: ± 1/16

Figure 2. Tread resistance test setup

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of 5.5 plus or minus 0.1. A test panel having a known resin system thickness shall be placed in a temperature chamber controlled at plus 94 plus or minus 3 degrees C (plus 200 plus or minus 5 degrees F) and allowed to remain at this temperature until stabilized. Within 5 minutes after removal from the temperature chamber, the test panel shall be exposed to the exhaust emanating from the cumbustion chamber for not less than 90 milliseconds. During exposure, the test panel shall be securely positioned, so that the surface of the resin system is downstream from and normal (plus or minus 5 degrees) to the flow from the combustion chamber nozzle and at a distance of 7 plus or minus 1 nozzle exit diameters from the nozzle exit. The test panel shall be allowed to cool to room temperature. Minimum thickness of the impinged area shall then be measured. Loss of resin system from the test panel shall be not more than 0.080 inch.

4.4.3.13 Shelf life. The base resin and curing agent shall be tested for conformance with all applicable requirements specified herein after completion of the 6 month storage period. The supplier's certificate of compliance with 3.3.13 may be accepted during the storage period.

4.4.3.14 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified in the contract or purchase order, the base resin and the curing agent shall be furnished separately in tinplated metal containers with comparable metal lids. Containers shall be non-reactive to the contents. Size of containers shall be specified by the procuring activity. The base resin and curing agent shall be packaged in such a manner that damage during shipment and storage is prevented as specified in MIL-P-14232.

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5.2 Marking. In addition to any special marking required by the contract or purchase order, such as shelf life, storage and cure date markings, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. Methyl-Ethyl-Ketone is a regulated item and shall be marked, packaged, and labelled in accordance with the hazardous material regulation applicable to the proposed mode of transportation.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as an ablative resistant deck coating and thermal insulation. Up to a resin thickness of 0.200 inch can be applied to vertical surfaces without subsequent flow.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required, and, if so, pertinent details (see 3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Requirements for preservation, packaging, packing, and marking, if different (see 5.1).

e. Size of containers (see 5.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS 16062, dated 6 December 1966.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 8030-A041

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